1 August 1986

MILITARY SPECIFICATION

IGNITER, MK 304 MOD 0

This specification is approved for use within the Naval Air Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the requirements for the manufacture and acceptance of the MK 304 Mod 0 Igniter (see 6.1).

2. APPLICABLE DOCUMENTS

2.1. Government documents.

2.1.1 Standards. The following standards form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

STANDARDS

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	Marking for Shipment and Storage
MIL-STD-331	Fuze and Fuze Components, Environmental and Performance Tests for
MIL-STD-414	Sampling Procedures and Tables for Inspection by Variables for Percent Defective
MIL-STD-453	Inspection, Radiographic
MIL-STD-810	Environmental Test Methods
MIL-STD-1167	Ammunition Data Card
MIL-STD-1168	Ammunition Lot Numbering
DOD-STD-2101	Classification of Characteristics

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commanding Officer, Naval Air Engineering Center, Systems Engineering and Standardization Department (SESD) Code 93, Lakehurst, NJ 08733, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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2.1.2 Drawings and publications. The following drawings and publications form a part of this specification to the extent specified herein. Unless otherwise specified, the issues shall be those in effect on the date of the solicitation.

DRAWINGS

Naval Air Systems Command (Code Ident 30003)

DL1638AS106	Igniter, Mk 304 Mod 0

Naval Ordnance Station, Indian Head, Maryland (Code Ident 14083)

DL560-174-001	Test Chamber, Mk 304 Mod 0

PUBLICATIONS

Code of Federal Regulations

49 CFR 100 - 199 -	Transportation

(Copies of specifications, standards, drawings and publications required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity).

2.2 Order of Precedence. In the event of a conflict between the text of this specification and the references cited herein (except for associated detail specifications, specification sheets or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified in the contract or order, a sample shall be subjected to first article inspection (see 4.4 and 6.3).

3.2 Conformance to documents. Igniters shall conform to all requirements of this document and DL1638AS106.

3.3 Radiographic examination (M101). All igniters shall be radiographically examined to determine the presence, location, and condition of components (see 4.6.2).

3.4 Circuit resistance 102). The igniter circuit resistance, as measured from the connector at 70° ± 10°F, shall be between 0.45 and 0.75 ohms (see 4.6.3).

3.5 Ballistic (M103). The igniters shall mat the performance requirements of table I over the temperature range of -40°F to 140°F (see 4.6.4).

TABLE 1. Ballistic performance requirements.

Characteristics 1/	Specification limits	
	Lower (L)	Upper (U)
Ignition delay (ms)	-	15
Maximum pressure (psig)	290	470
Rise time (ms)	-	11

1/ See 6.4 for definitions.

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3.6 Workmanship. Igniters and components shall be free from foreign material, dents, scratches, split or cracked components, poor quality welds, chipping or flaking of painted metal surfaces, sharp edges or other defects or discontinuities as specified on the applicable drawings.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may usc his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Governmentt to acceptance of defective material.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.4)
b. Quality conformance inspections (see 4.5)

4.3 Inspection conditions. Unless otherwise specified (see 6.2.1), all inspections shall be performed as specified in the applicable test method of 4.4.3 or 4.6.

4.4 First article inspection. First article inspection shall be performed after award of contract and prior to production by a facility designated in the contract or order (see 6.2.1 and 6.2.2). First article inspection shall be performed on sample units which have been produced using the same materials, equipment, processes and procedures as will be used in production. All parts and materials, including packaging and packing shall be the same as used for production and shall be from the intended sources of supply.

4.4.1 Sample size. Twenty-four igniters shall be submitted for first article inspection.

4.4.2 Inspection routine. The igniters shall undergo the tests specified in table II in the exact sequence shown.

4.4.3 Test and treatment.

4.4.3.1 Vibration. The igniter per Drawing 1638AS106 shall be subjected to Category I, Basic Transportation Vibration, method 514.3 schedules defined by Figures 514.3-1 through 514.3-3 of MIL-STD-810. Expected transportation distance shall be 3000 miles and the igniter shall be treated as secured cargo. The igniter shall be maintained at the temperature specified in Table II.

4.4.3.2 Temperature and humidity. Igniters shall be subjected to the temperature and humidity test of MIL-STD-331, Test 105.1, except that the temperature extremes shall be - 40°F and + 140°F and only one 14-day cycle shall be performed.

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TABLE II. First article inspection 1/.

Inspection and test	Igniter serial number																								Notes
	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	
Visual	a	a	a	a	a	a	a	a	a	a	a	a	a	a	a	a	a	a	a	a	a	a	a	a	see 4.6.1
Circuit resistance	b	b	b	b	b	b	b	b	b	b	b	b	b	b	b	b	b	b	b	b	b	b	b	b	see 4.6.3
Radiographic	c	c	c	c	c	c	c	c	c	c	c	c	c	c	c	c	c	c	c	c	c	c	c	c	see 4.6.2
Vibration at −40°F													d	d											see 4.4.3.1
Vibration at 140°F															d	d									
Temp. and humidity																	d	d	d	d					see 4.4.3.2
2-foot drop at −40°F																					d		d		see 4.4.3.3
2-foot drop at 140°F																						d		d	
Circuit resistance	e	e	e	e	e	e	e	e	e	e	e	e	e	e	e	e	e	e	e	e	e	e	e	e	see 4.6.3
Radiographic													f	f	f	f	f	f	f	f	f	f	f	f	see 4.6.2
Static fire at −40°F	d	d	d	d	d	d							g			g	g	g			g			g	see 4.6.4
Static fire at 140°F							d	d	d	d	d	d		g	g				g	g		g	g	

1/ Inspection and tests of each igniter shall be performed in alphabetical order; i.e., the inspection or test opposite the letter "a" shall be performed first, the inspection or test opposite the letter "b" shall be performed second, etc.

4.4.3.3 Drop test. The igniters shall be dropped from 2 feet, measured from the lowest point on the igniter to the impact surface. The impact surface shall be a 2-inch thick steel plate (Brinell hardness of not less than 207) embedded in not less than of 2 feet of reinforced concrete. Two igniters shall be temperature conditioned and maintained at − 40°F and two at 140°F. Of these, two (one from each temperature) shall be dropped on the bottom surface of the igniter plug and two (one from each temperature) dropped on the flange of the igniter case.

4.4.4 Failures. Failure of any igniter to pass the first article tests shall be cause for first article rejection

4.5 Quality conformance inspection. Quality conformance inspection shall consist of the tests specified in table III (see 6.2.2).

TABLE III. Quality conformance inspection.

Inspection or test	Req. para.	Method para.
Visual	3.6	4.6.1
Radiographic	3.3	4.6.2
Circuit resistance	3.4	4.6.3
Ballistic	3.5	4.6.4

4.5.1 Inspection lot. An inspection lot shall consist of not more than 1000 igniters presented for acceptance at one time. No more than one lot of magnesium-fluorocarbon and one lot of initiators shall be used in an igniter lot.

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4.5.2 Sampling.

4.5.2.1 For nondestructive tests. Unless otherwise specified in the applicable test method, sampling for nondestructive inspection shall be in accordance with MIL-STD-105, inspection level II, single sampling plans for normal inspection, with an acceptable quality level (AQL) of 1.0 percent for major characteristics and 2.5 percent for minor characteristics.

4.5.2.2 For destructive tests. The ballistic acceptance sample for the igniter lot shall be selected and evaluated in accordance with MIL-STD-414, inspection level III, variabilty unknown, standard deviation method, with an AQL of 1.0 percent. In the event of static-firing failure, the sample firings will be terminated for the remaining sample units pending disposition instructions from the procurring activity.

4.5.3 Classification of characteristics. The characteristics verified by the tests and examinations herein are classified as critical, major, or minor in accordance with DOD-STD-2101. Requirements that verify critical characteristics are identified by the symbol (C) and major characteristics by the symbol (M). The number following the classification symbol indicates the serial number of the requirement. Requirements which are not annotated with a classification code are classified minor.

4.6 Tests.

4.6.1 Visual. The igniter samples shall be visually examined for conformance to DL1638AS106.

4.6.2 Radiographic. The igniters shall be radiographer in accordance with MIL-STD-453, quality level 1 for conformance to DL1638AS106.

4.6.3 Circuit resistance. The igniters shall be checked for circuit resistance by applying 0.010 amperes (maximum) to the initiator assembly circuit at 70° ± 10°F.

4.6.4 Ballistic test. The igniter samples selected in accordance with 4.5.2.2 shall be static fired (within 10 minutes of removal from conditioning) as specified in 4.6.4.1 through 4.6.4.4.

4.6.4.1 Temperature conditioning. Half the igniters shall be conditioned at - 40°F and the other half at 140°F to assure a uniform temperature throughout the entire mass at the time of test. The conditioning time, at either temperature, shall be not less than 8 hours. Igniters not fired within 10 minutes after removal from the conditioning oven shall be reconditioned 2 minutes for each minute (over the 10) outside the conditioning chamber. The reconditioning time shall be not more than 8 hours.

4.6.4.2 Test setup. Igniters shall be fired in a closed chamber conforming to DL560-174-001. Pressure shall be measured through the pressure ports of the closed chamber. The igniters shall be fired by energizing the igniter circuit with 28 volts dc at 7.0 + 0.5, -0.0 amperes.

4.6.4.3 Instrumentation. All measurement shall be made with instruments and systems With a total system accuracy, reproducibility, and fequency response that has been approved by the procuring acitivity. Calibration records of all test equipment shall be maintained and made available to the procuring activity upon request.

5. PACKAGING

5.1 Packaging and packing. The igniter shall be packaged and packed in accordance with 49 CFR 171-179 or as specified in the purchase order, requisition, or contract (see 6.2.1).

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5.2 Marking. In addition to any special marking required by the contractor order (see 6.2.1), unit packages and shipping containers shall be marked in accordance with MIL-STD-129 and 49 CFR 171-179. All information appearing on the igniter package shall be stenciled or otherwise marked on the exterior of the igniter package.

5.3 Ammunition data cards. Ammunition data cards, completed in accordance with MIL-STD-1167, shall be included with each lot (see 6.2.2).

5.4 Igniter lot numbering system. Lot numbering shall be as specified in MIL-STD-1168.

6. NOTES

6.1 Intended use. The Mk 304 Mod 0 igniter is intended for use as the ignition device for the Rocket Motor Mk 117 Mod 0.

6.2 Ordering data.

6.2.1 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number and date of this specification
b. Whether first article inspection is required and if so, specify the test activity (see 4.4)
c. Production lot size and test activity (see 4.5. i)
d. That the safety precaution requirements of the "Contractors' Safety Manual for Ammunition, Explosives, and Related Dangerous Material," DOD 4 145.26M, are applicable
e. Inspection conditions, if other than as specified (see 4.3)
f. Packing and marking requirements, if other than as specified (see 5.1 and 5.2)

6.2.2 Data requirements. When this specification is used in an acquisition and data are required to be delivered, the data requirements identified below shall be developed as specified by an approved Data Item Description (DD Form 1664) and delivered in accordance with the approved Contract Data Requirements List (CDRL), incorporated into the contract. When the provisions of DoD FAR Supplement, Part 27, Sub-Part 27.410-6 (DD Form 1423) are invoked and the DD Form 1423 is not used, the data specified below shall be delivered by the contractor in accordance with the contract or purchase order requirements. Deliverable data required by this specification are cited in the following paragraphs.

Paragraph no.	Data requirement title	Applicable DID no.	Option
4.4	First article inspection procedure	DI-T-4901	-
4.4	First article inspection reports	DI-T-4902	-
4.5	Production acceptance inspection procedures	DI-T-4903	-
4.5	Production inspection reports	DI-T-4904	-
5.3	Ammunition Data Card	DI-MISC-80043	-

(Data item descriptions related to this specification, and identified in this section will be approved and listed as such in DoD 5000.19L., Vol.II, AMSDL. Copies of data item descriptions required by the contractors in connection with specific acquisition functions should be obtained from the Naval Publications and Forms Center or as directed by the contracting officer.)

6.3 First article. When a first article inspection is required, the item should be a first article sample. The first article should consist of 24 igniters. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, approval of first article test results and disposition of first articles. Invitations for bids should provide that the Government reserves the right to waive

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the requirements for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract.

6.4 Definitions. For the purpose of this specification, the following definitions apply.

6.4.1 Ignition delay. Ignition delay is the time from application of firing current to first indication of pressure on the pressure-time trace, expressed in milliseconds.

6.4.2 Maximum pressure. Maximum pressure is the highest pressure at any point of the pressure-time trace, expressed in psig.

6.4.3 Rise time. Rise time is the time from first indication of pressure to the time pressure reaches 140 psig on the pressure-time trace, expressed in milliseconds.

6.5 Subject term (key word) listing:

Igniter, MK 304
Test chamber, MK 304
Ballistic (M102)
Circuit resistance (M102)

Preparing activity:
Navy - AS

(Project 1340-N713)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

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1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-P-85760(AS)	IGNITER, MK 304 MOD 0

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*: _____
b. ADDRESS *(Street, City, State, ZIP Code)*	

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording.

c. Reason/Rationale for Recommendation

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
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